A FINITE TYPE MODULE OVER AN AFFINE FINITE TYPE SCHEME IS PUSHED FORWARD FROM ITS SUPPORT BY A FINITELY PRESENTED CLOSED IMMERSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. All four referenced results live in earlier chapters, so each locator points at the verbatim line of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carries the canonical tag.

Allowed standard prerequisites (editorial).

- Support of a finite type module is the closed subscheme cut out by its annihilator. Algebra, Lemma labelled algebra-lemma-support-closed (canonical Stacks tag 00L2); invoked at proof line 468. Supplies the description of Supp(F) via the annihilator ideal I of M; the comparison of this description with the fibre is the counted step.
- Characterisation of a finite type morphism of affine schemes as a finite type ring map. Morphisms, Lemma morphisms-lemma-locally-finite-type-characterize (tag 01T2, line 461) and Properties, Lemma properties-lemma-finite-type-module (tag 01PB, line 463). Supply that A -> B is a finite type ring map and that M is a finite B-module, which is what allows finitely many annihilator elements to be chosen later.
- Support of a finite type module on a fibre. Morphisms, Lemma labelled morphisms-lemma-support-finite-type (canonical Stacks tag 056J); invoked at proof line 474. Supplies that Supp(F_s) = V(I(B tensor_A kappa(p))); the choice of f_1, ..., f_m cutting out that closed set is inside the counted span.

Source reference. Original lemma 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Cross-chapter standard prerequisite invoked once at proof line 461 to turn the finite type morphism of affines into a finite type ring map; it is a general equivalence and replaces nothing in the retained argument.

Source reference. Original lemma 01PB: sources/stacks-chow/tags.txt, source lines 2224–2224. Cross-chapter standard prerequisite invoked once at proof line 463 to know that M is a finite B-module; the choice of the annihilator generators is the counted step and stays in the delivered body.

Source reference. Original lemma 00L2: sources/stacks-chow/tags.txt, source lines 772–772. Cross-chapter standard prerequisite invoked once at proof line 468 for the annihilator description of the support of a finite module.

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Source reference. Original lemma 056J: sources/stacks-chow/tags.txt, source lines 6712–6712. Cross-chapter standard prerequisite invoked once at proof line 474 for the description of the support of the restriction of a finite type module to a fibre.

Lemma 0.1. *Let $f : X \to S$ be a finite type morphism of affine schemes. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module. Let $x \in X$ with image $s = f(x)$ in S. Set $\mathcal{F}_s = \mathcal{F}|_{X_s}$. Then there exist a closed immersion $i : Z \to X$ of finite presentation, and a quasi-coherent finite type $\mathcal{O}_Z$-module $\mathcal{G}$ such that $i_*\mathcal{G} = \mathcal{F}$ and $Z_s = Supp(\mathcal{F}_s)$.*

Proof. Say the morphism $f : X \to S$ is given by the ring map $A \to B$ and that $\mathcal{F}$ is the quasi-coherent sheaf associated to the B-module M. By Morphisms, Lemma 01T2 we know that $A \to B$ is a finite type ring map, and by Properties, Lemma 01PB we know that M is a finite B-module. In particular the support of $\mathcal{F}$ is the closed subscheme of $\mathrm{Spec}(B)$ cut out by the annihilator $I = \{x \in B \mid xm = 0 \ \forall m \in M\}$ of M, see Algebra, Lemma 00L2. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to x and let $\mathfrak{p} \subset A$ be the prime ideal corresponding to s. Note that $X_s = \mathrm{Spec}(B \otimes_A \kappa(\mathfrak{p}))$ and that $\mathcal{F}_s$ is the quasi-coherent sheaf associated to the $B \otimes_A \kappa(\mathfrak{p})$ module $M \otimes_A \kappa(\mathfrak{p})$. By Morphisms, Lemma 056J the support of $\mathcal{F}_s$ is equal to $V(I(B \otimes_A \kappa(\mathfrak{p})))$. Since $B \otimes_A \kappa(\mathfrak{p})$ is of finite type over $\kappa(\mathfrak{p})$ there exist finitely many elements $f_1, \ldots, f_m \in I$ such that

$$I(B \otimes_A \kappa(\mathfrak{p})) = (f_1, \ldots, f_n)(B \otimes_A \kappa(\mathfrak{p})).$$

Denote $i : Z \to X$ the closed subscheme cut out by $(f_1, \ldots, f_m)$, in a formula $Z = \mathrm{Spec}(B/(f_1, \ldots, f_m))$. Since M is annihilated by I we can think of M as an $B/(f_1, \ldots, f_m)$-module. In other words, $\mathcal{F}$ is the pushforward of a finite type module on Z. As $Z_s = \mathrm{Supp}(\mathcal{F}_s)$ by construction, this proves the lemma. $\square$